

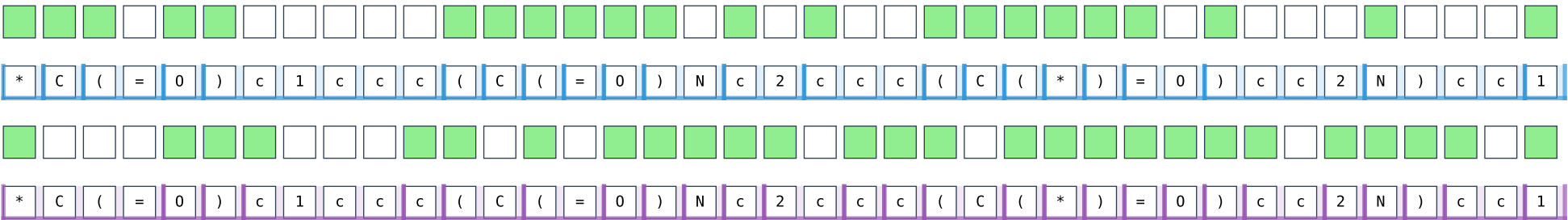
Evolution of H-Net Tokenization During Training

Example PSMILES: *C(=O)c1ccc(C(=O)Nc2ccc(C(*)=O)cc2N)cc1

Green = boundary marker | Blue brackets = 1-Stage tokens | Purple brackets = 2-Stage tokens

1-Stage (22 tokens, avg 1.8 chars)

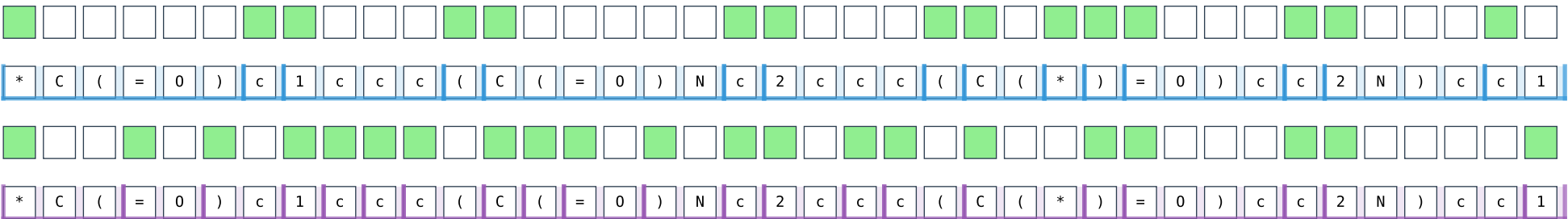
Early (1M bytes)



2-Stage (27 tokens, avg 1.4 chars)

1-Stage (15 tokens, avg 2.6 chars)

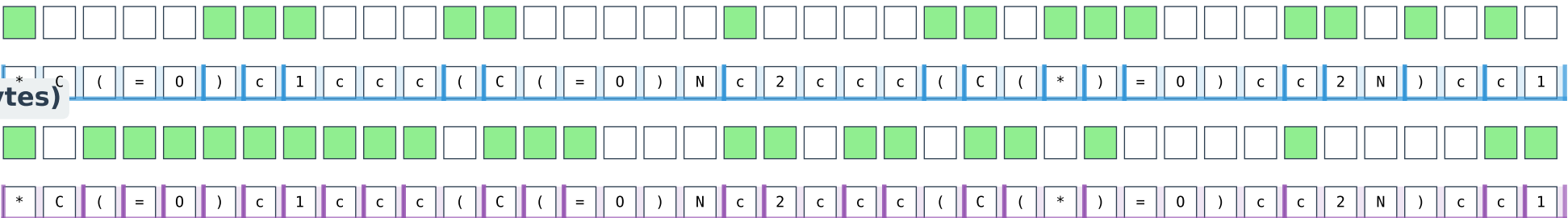
Mid (100M bytes)



2-Stage (21 tokens, avg 1.9 chars)

1-Stage (16 tokens, avg 2.4 chars)

Converged (238M bytes)



2-Stage (23 tokens, avg 1.7 chars)